

Co-Pyrolysis of Coal and Waste Plastic: Characterization and Rapid Tar Yield Prediction Method

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Abstract: Waste plastics, with their high hydrogen-to-carbon (H/C) atomic ratios, can act as hydrogen donors during coal pyrolysis, thereby enhancing tar yield and quality. Thus far, a study has been conducted on the co-pyrolysis characteristics of coal and waste plastic, along with a rapid prediction method for tar yield. An experimental system for the co-pyrolysis of coal and waste plastic is established to examine the distribution patterns of pyrolysis products, such as gas, tar, and char, at varying temperatures and coal-to-waste plastic ratios. The results indicate a significant synergistic effect during the co-pyrolysis of coal and plastic waste. As the blending ratio of waste plastic increases, the tar yield also increases, with the value of the synergistic effect parameter initially rising and then falling. As the blending ratio continues to increase, the formation of a liquid phase becomes more prevalent on the surface of coal particles during the pyrolysis process, which inhibits tar release and leads to a gradual decrease in the positive synergistic effect of the waste plastic on tar yield. Based on these findings, a rapid prediction model for tar yield has been developed using neural networks and optimized with a Genetic Algorithm (GA) and Particle Swarm Optimization (PSO), achieving a 10.52% reduction in the average prediction error under training conditions. The proposed model is utilized to predict the tar yield for new conditions in the database, with the relative error generally maintained within (–20%, 30%), demonstrating good accuracy and utility.

Keywords: coal; waste plastic; co-pyrolysis; tar yield; rapid prediction

1. Introduction

Coal is China's primary energy source, with its production and consumption dominating the primary energy sector [1]. Low-rank coal, which constitutes over 50% of coal reserves and output, leads to significant resource wastage and environmental pollution when used directly [2, 3]. Utilizing low-rank coal through pyrolysis

technology represents an effective technical route for the clean and efficient transformation of coal resources [4–6]. However, coal tar, a valuable high-value-added product of this technology, has a low yield when only low-rank coal is pyrolyzed due to its low hydrogen-to-carbon (H/C) atomic ratio.

Plastic is a kind of high molecular weight hydrocarbon polymer; due to the advantages of low density, high

plasticity and strong corrosion resistance, it is widely used in industry, agriculture, construction and other fields. Plastic for various applications is increasing every year; but after being used, the recycling and disposal have remained a challenge [7]. Over 75% of waste plastics ultimately end up being landfilled, incinerated, or entering the natural environment, which not only wastes valuable resources but also seriously threatens the environment and human health due to the dust and dioxins produced by combustion [8]. Waste plastic disposal presents a significant opportunity for energy recovery. Pyrolysis is an environmentally friendly and cost-effective technology for recovering energy from waste plastic by converting waste plastic into basic petrochemicals [9, 10]. Mangesh et al. [11, 12] used a Ni-Mo/Laponite catalyst to produce hydro-processed waste polypropylene pyrolysis oil, whose physicochemical properties and performance match those of premium commercial diesel. Pyrolytic oil extraction from waste plastic alone is a resourceful recycling method; but it has drawbacks such as small scale and inability to handle chlorine-containing waste plastic, which prevents the large-scale application of waste plastic pyrolysis [13].

Currently, the co-pyrolysis of coal and waste plastics has garnered widespread attention. Existing research indicates that waste plastic can serve as an effective hydrogen donor during coal pyrolysis, with co-pyrolysis exhibiting a notable positive synergistic effect that significantly increases tar yield [14]. Zhou et al. [15] employed thermogravimetric analysis to study the pyrolysis behavior of low-volatility coal, plastic, and their mixtures. They observed an overlapping pyrolysis temperature range and a positive synergistic effect at higher temperatures. Melendi-Espina et al. [16] used thermogravimetric mass spectrometry (TG-MS) to study the co-pyrolysis behavior of coking coal mixtures with various plastics under an N₂ atmosphere. They identified a synergistic effect between the volatiles released during coal pyrolysis and plastic, with maximum interaction occurring near the peak release temperature of the plastic's volatile components. Zhang et al. [17] analyzed the co-pyrolysis products of pre-dechlorinated polyvinyl chloride (PVC) residue (DPVC) and Pingshuo coal, noting a significant positive synergistic effect on the generation of coal tar and char, and a negative synergistic effect on the formation of pyrolysis water and gas. An addition of 10% DPVC resulted in a 5% increase in light tar content compared to the calculated value. The co-pyrolysis characteristics of coal and plastic are primarily influenced by the pyrolysis feedstock and the thermal reaction conditions. Havelcová et al. [18] investigated the impact of blending ratios on the co-pyrolysis yields. The yield of tar increases with the

addition of PET (Polyethylene Terephthalate), and then decreases when the proportion of PET is higher than 20%. Sinag et al. [19] conducted research on the co-pyrolysis of lignite and low-density polyethylene, and found that as the final pyrolysis temperature increases, the yield of tar first increases and then decreases. The reasons for these changes are the enhanced stability of free radicals and the suppression of cross-linking structures. Liao et al. [20] found that using coke oven gas instead of pure hydrogen for pyrolysis can increase tar yield and is more cost-effective, offering better economic feasibility.

Using tar yield as a quantitative index of the synergistic effect can effectively guide the application of co-pyrolysis technology [21]. Therefore, obtaining tar yields under various operating conditions enables the exploration of optimal pyrolysis feedstocks and conditions, offering valuable insights for developing technologies that produce high-quality coal tar through the co-pyrolysis of coal and waste plastic. However, due to the complexity of raw material types and pyrolysis conditions in actual experiments, the time and cost required to determine coal tar yield are relatively high. With technological advancements, machine learning methods have found broader applications. The BP neural network, a well-established neural network model, can rapidly analyze complex nonlinear relationships from basic experimental data and construct a prediction model for products [22]. Ahmad et al. [23] provided three machine learning-based approaches, i.e., Gradient boosting regression (GBR), extreme gradient boosting (XGB), and light gradient boosting machine (LGBM) regression, for prediction and genetic algorithm for the optimization of biodiesel yield from waste cooking oil. The findings have suggested that the LGBM model possesses excellent predictive capabilities and can assist in making informed decisions regarding biodiesel yield prediction. Qiao et al. [24] constructed two tar yield prediction models employing the BP neural network, based on 20 coal quality indices and 4 proximate analytical indices respectively. The evaluation results have shown that the first model features higher goodness of fit and generalization performance. Nevertheless, the simplicity of the BP neural network implies that it is an unconstrained nonlinear optimization process, which can result in low search efficiency and a tendency to converge to local optima, potentially leading to suboptimal results [25].

In light of the above literature, the co-pyrolysis of coal and waste plastic is theoretically feasible, and the synergistic effect can increase the yield of tar. However, there is currently a lack of quantitative indicators for the synergistic effect, which cannot yet guide the application of co-pyrolysis oil production technology. On the other hand, obtaining tar yield through experiments is costly

and time-consuming. With the development and improvement of neural network technology, some scholars have applied it to the prediction of pyrolysis products of plastic and/or biomass [26–29] and achieved good predictive results. However, there are few reports on neural network models for the prediction of tar yield of coal and waste plastic co-pyrolysis. This study first conducted experiments on the co-pyrolysis characteristics of coal and waste plastic. The focus is on examining the impact of different pyrolysis feedstocks and conditions on the synergistic effects and distribution patterns of pyrolysis products, while also obtaining the tar yield under various conditions and determining the input and output parameters for the neural network structure. Subsequently, a BP neural network model for the rapid prediction of tar yield is constructed and trained using both experimental and literature data [30–33]. In consideration of the limitations of the BP neural network prediction model, the model is optimized with a Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) to identify the most effective prediction model for tar yield. The accuracy of this optimized prediction model is then verified using database conditions.

2. Materials and Methods

2.1 Experimental materials

Bituminous coal, lignitous coal, and waste plastic are selected as the research subjects. The preliminary testing of the raw materials includes proximate and ultimate analyses, with the results presented in Table 1. The coal and waste plastic are crushed and sieved to a particle size range of 0.2–0.45 mm, and then dried in a drying oven at 105°C for 12 h for later use. The dried coal and waste plastic are mechanically mixed in different proportions. The samples are labeled as “Coal + *x* Waste Plastic”, where *x* represents the blending ratio of waste plastic in the sample, with *x*=0%, 20%, 40%, 60%, 80%, and 100%.

2.2 Experimental apparatus and procedure

The experimental apparatus for the co-pyrolysis of coal and waste plastic consist mainly of a gas distribution system, heating furnace, thermoregulator, insulation device, and tar condensation collection device, as shown in Fig. 1. Initially, the optimal pyrolysis temperature is determined based on the tar yield from coal pyrolysis alone. Co-pyrolysis experiments with coal and waste

Table 1 Proximate and ultimate analysis of samples

Sample	Proximate analysis for mass percentage ^{ar} /%				Ultimate analysis for mass percentage ^{daf} /%				
	Moisture (M)	Ash (A)	Volatile (V)	Fixed carbon (FC)	C	H	O	N	S
Bituminous coal	4.51	10.79	31.24	53.46	62.09	3.663	33.467	0.66	0.12
Lignitous coal	14.09	11.93	44.51	29.47	64.99	5.367	27.673	0.90	1.07
Waste plastic	0.29	0.02	93.28	6.41	55.42	6.94	37.53	0	0.11

Note: ar, on an air received basis; daf, on a dry ash free basis. O content is calculated by difference.

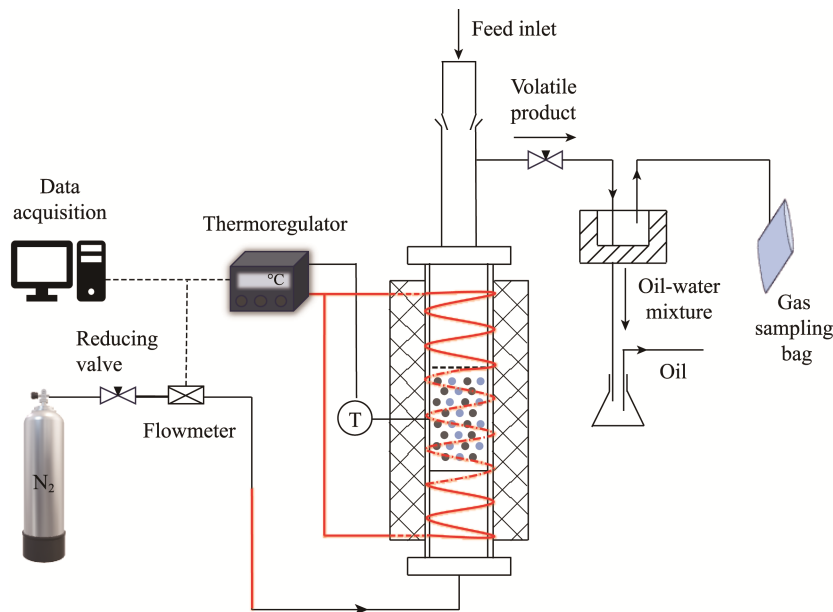


Fig. 1 Schematic of the co-pyrolysis experimental setup for coal and waste plastic

plastic are then conducted at this optimal temperature. During the experiment, ten grams of uniformly mixed samples are placed in the upper basket of the reactor under a continuous flow of N_2 (300 mL/min) as the carrier gas. Once the heating device reaches and stabilizes at the designated temperature, the sample is quickly added to the reaction tube within 1 minute, and the pyrolysis is maintained for 40 minutes before heating ceasing. The experiment concludes after an additional 30 minutes of gas sweeping. The solid yield (char) is calculated by the mass difference between the post-pyrolysis and empty basket. Liquid products (tar and water) are collected and weighed using a cold trap at -10°C . Pyrolysis water is separated from the oil-water mixture according to the standard GB/T 480-2010, and the water content is recorded using an oil-water separator. Tar and gas yields are calculated by difference.

The char yield (Y_{char}), tar yield (Y_{tar}), pyrolysis water yield (Y_{water}), and gas yield (Y_{gas}) are calculated using the following formulae, with all calculations based on a dry-ash-free basis:

$$Y_{\text{char}} = \frac{W_{\text{char}} - \sum_{i=1}^2 W_0 x_i A_{t,i}}{\sum_{i=1}^2 W_0 x_i (1 - A_{t,i} - M_{t,i})} \times 100\% \quad (1)$$

$$Y_{\text{tar}} = \frac{W_{\text{tar}}}{\sum_{i=1}^2 W_0 x_i (1 - A_{t,i} - M_{t,i})} \times 100\% \quad (2)$$

$$Y_{\text{water}} = \frac{W_{\text{water}} - \sum_{i=1}^2 W_0 x_i A_{t,i}}{\sum_{i=1}^2 W_0 x_i (1 - A_{t,i} - M_{t,i})} \times 100\% \quad (3)$$

$$Y_{\text{gas}} = 100\% - Y_{\text{char}} - Y_{\text{tar}} - Y_{\text{water}} \quad (4)$$

where W_{char} , W_{water} , W_{tar} , and W_0 represent the masses of char, water, tar, and the sample, respectively; $A_{t,i}$ and $M_{t,i}$ denote the ash and moisture percentage content of the dried sample, respectively, and x_i is the mass fraction of coal and waste plastic in the sample.

The synergistic effect of the co-pyrolysis of coal and waste plastic is assessed using the synergistic effect parameter Y_{SE} , which can be calculated as follows [34]:

$$Y_{\text{cal}} = x_1 Y_1 + x_2 Y_2 \quad (5)$$

$$Y_{\text{SE}} = \frac{Y_{\text{exp}} - Y_{\text{cal}}}{Y_{\text{cal}}} \times 100\% \quad (6)$$

where Y_1 and Y_2 are the experimental tar yields from the individual pyrolysis of coal and waste plastic, respectively, and Y_{cal} and Y_{exp} are the calculated and experimental values of the tar yield for the co-pyrolysis of coal and waste plastic, respectively. Tar yield is used as the evaluation index of synergistic effect parameter, which is the main concern of this paper.

2.3 BP prediction model for tar yield

2.3.1 Neural network topology

The BP neural network is a multilayer feedforward neural network designed based on back propagation of errors, consisting primarily of three layers: the input layer, hidden layer, and output layer. The number of hidden layers can impact the structure and predictive accuracy of the network. Theoretically, a BP neural network with a single hidden layer is capable of approximating any complex nonlinear function, offering a simple network structure and rapid fitting. Consequently, this study employs a three-layer BP neural network to develop a rapid prediction model for tar yield, as illustrated in Fig. 2.

Based on the co-pyrolysis experiments, the input parameters are defined as the percentage of composition of coal and waste plastic (i.e. A, V, FC, C, H, O) and the

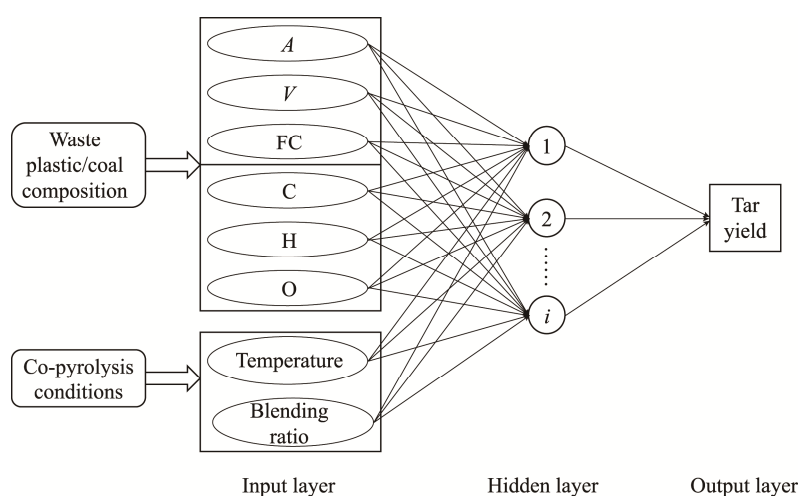


Fig. 2 Schematic of neural network topology

co-pyrolysis conditions (temperature and blending ratio), with 14 nodes in the input layer. The output parameter is tar yield, represented by a single node in the output layer. The number of nodes in the hidden layer is initially estimated using an empirical formula and subsequently refined through repeated calculations. The empirical formula [35] used is:

$$l < \sqrt{m+n} + a \quad (7)$$

where l is the number of neurons in the hidden layer; m is the number of neurons in the input layer; n is the number of neurons in the output layer, and a is a constant between 0 and 10.

2.3.2 Data preprocessing

In most practical applications of BP neural network prediction models, the initial input values can cause extreme polarization of neuron weights, leading to prediction errors. Therefore, it is common practice to normalize the input data to a range between 0 and 1. This normalization is achieved using the following formula [36–38]:

$$\bar{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (8)$$

where, x_i is the input data; $x_{\min}$ represents the minimum value within the data, and $x_{\max}$ denotes the maximum value.

2.3.3 Activation functions

Although the BP neural network is considered one of the most successful neural network learning algorithms due to its ability to perform complex nonlinear mappings, it has certain limitations. These include slow or oscillatory convergence, a tendency to become trapped in local minima, and a risk of overfitting. To mitigate these issues in practical applications, it is common to adjust the learning rate; select appropriate activation functions, and combine the BP neural network with other optimization methods. In this study, a database containing 140 data points is created using experimental and literature data [35–38], and various optimization techniques are applied to refine the model and determine the optimal rapid prediction model for tar yield.

The incorporation of activation functions into neural network models introduces nonlinearity, thereby enhancing the network's learning capabilities. However, traditional activation functions such as tansig, logsig, and purelin can enter saturation regions during computation, which may adversely affect the model's output [39]. This study introduces the Elliotsig and Satlins functions and evaluates 25 combinations of these five activation functions to determine the optimal configuration. The specific expressions for the aforementioned activation functions are as follows:

$$\text{tansig}(x) = \frac{2}{1 + e^{-2x}} - 1 \quad (9)$$

$$\text{logsig}(x) = \frac{1}{1 + e^{-x}} \quad (10)$$

$$\text{purelin}(x) = x \quad (11)$$

$$\text{elliotsig}(x) = \frac{x}{1 + |x|} \quad (12)$$

$$\text{satlins}(x) = \begin{cases} -1 & x < -1 \\ x & x \in [-1, 1] \\ 1 & x > 1 \end{cases} \quad (13)$$

Seven sets of conditions are selected as the training set, including Pingshuo coal mixed with 0%, 50%, and 100% waste plastic WPE, as well as Pingshuo coal mixed with 0%, 30%, 50%, and 100% waste plastic BPE. Additionally, three sets of conditions are chosen as the testing set, including Pingshuo coal mixed with 10% and 30% waste plastic WPE, and Pingshuo coal mixed with 10% waste plastic BPE. The neural network model consists of an input layer with 14 nodes, a hidden layer with four nodes, and an output layer with one node. The performance of the model is assessed using the mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2), with smaller prediction errors and

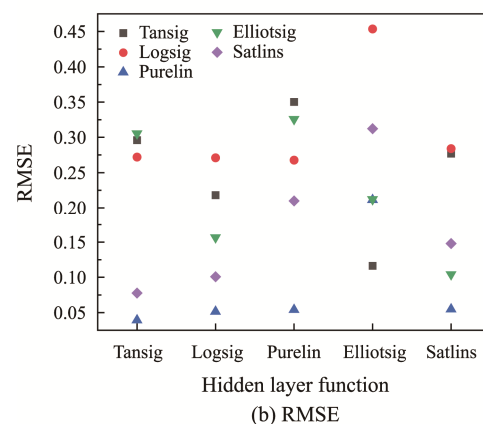
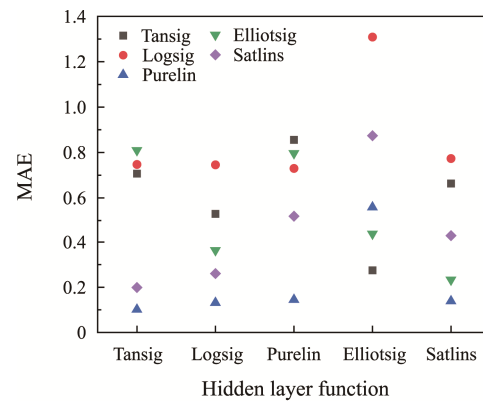


Fig. 3 Prediction error of different activation function combinations

higher R^2 values indicating higher model accuracy. The formulas [40] are as follows:

$$\text{MAE} = \frac{1}{N} \left(\sum_{i=1}^N |H_i - H_{pi}| \right) \quad (14)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \left(\sum_{i=1}^N |H_i - H_{pi}|^2 \right)} \quad (15)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{H_i - H_{pi}}{H_i} \right| \times 100\% \quad (16)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (H_i - H_{pi})^2}{\sum_{i=1}^N (H_i - \bar{H})^2} \quad (17)$$

where H_i represents the test data; H_{pi} denotes the predicted data; $\bar{H}$ indicates the mean of the test data, and N refers to the number of tests.

Each combination of activation functions is used for three separate predictions, with the average value considered as the final prediction error. As illustrated in Fig. 3, the model achieves superior accuracy when the output layer employs the linear purelin function and the hidden layer uses the nonlinear tansig, logsig, and satlins functions, resulting in errors with an order of magnitude smaller compared to other combinations. Specifically, when the hidden layer utilizes the tansig function and the output layer uses the purelin function, the model achieves the highest prediction accuracy.

3. Results and Discussion

3.1 Influence of temperature on the distribution of products from individual pyrolysis of coal and waste plastic

Table 2 presents the tar yields from the individual pyrolysis of bituminous coal, lignitous coal, and waste plastic under atmospheric pressure, with pyrolysis temperatures ranging from 500°C to 700°C and a carrier gas flow rate of 300 mL/min. The results indicate that the tar yields from the individual pyrolysis of the two coals and waste plastic increase initially with rising pyrolysis temperatures, then decrease beyond a certain point. For bituminous and lignitous coals, the optimal pyrolysis temperature is 550°C. This suggests a significant influence of temperature on the tar yield during coal pyrolysis. The increase of temperature promotes the depolymerization and decomposition of volatiles, resulting in an enhanced release of volatile species, which become gas and tar after cooling. When the temperature rises to a certain extent, while continuing to promote the generation of volatiles, there is a concurrent enhancement of secondary reactions involving these

species. Such secondary reactions include thermal cracking, polymerization, and carbonization, which lead to the conversion of tar components into gas and solid products, thereby reducing the tar yield [41].

Additionally, within the temperature range of 500°C to 600°C, the tar yield from lignitous coal is higher than that from bituminous coal, likely due to lignitous coal's higher volatile content. The tar yield from plastic waste pyrolysis is significantly higher than that from coal, which may be attributed to the greater volatile content and thermal decomposability of waste plastic.

The tar yield from plastic pyrolysis is less sensitive to temperature within the indicated range, which is mainly due to the differences in chemical structure and pyrolysis characteristics between the two. Coal is a complex organic ore whose pyrolysis process involves complex scission and reconstitution of chemical bonds. In contrast, plastic is composed of relatively homogeneous polymer compounds; its molecular structure is simpler than coal, and the chemical reaction pathways during pyrolysis are relatively direct. In addition, the pyrolysis of plastics generally occurs within a narrower temperature range than coal [21], without the intense secondary reactions that characterize coal pyrolysis.

Table 2 Tar yield from individual pyrolysis of raw materials

Coal type	Tar yield at different temperature/%				
	500°C	550°C	600°C	650°C	700°C
Bituminous coal	4.50	6.35	5.75	5.55	3.65
Lignitous coal	6.76	7.87	6.05	4.97	1.96
Waste plastic	59.90	61.16	62.19	61.82	60.74

3.2 Influence of blending ratio on the distribution of co-pyrolysis products of coal and waste plastic

Fig. 4 illustrates the effect of blending ratio on the distribution of co-pyrolysis products and the synergistic effect parameter values for bituminous coal and waste plastic under atmospheric pressure, with a pyrolysis temperature of 550°C, a heating rate of 10°C/min, and a carrier gas flow rate of 300 mL/min. As shown in Fig. 4(a), the yields of char and pyrolysis water gradually decrease with increasing blending ratios of waste plastic, while the yields of tar and pyrolysis gas increase. Fig. 4(b) indicates that the addition of waste plastic to bituminous coal results in a positive synergistic effect on tar formation, with the synergistic effect parameter value first increasing and then decreasing as the blending ratio rises, reaching a maximum value of 9.89% at a 40% blending ratio. This suggests that the presence of plastic waste promotes the combination of volatile free radicals [42]. As the blending ratio increases, the formation of a liquid phase during pyrolysis becomes more prevalent,

covering the coal particles and inhibiting tar release, which leads to a decrease in the synergistic effect parameter value. The yield of pyrolysis water is lower than the calculated value, indicating that co-pyrolysis suppresses water generation. Furthermore, the addition of waste plastic initially exerts a negative synergistic effect on the yield of co-pyrolysis gas, as the molten plastic tends to cover the coal particles' surface, thereby limiting gas release. However, as the plastic waste content increases, the negative synergistic effect gradually diminishes. This may be due to the higher plastic waste content releasing a substantial amount of volatile substances, which helps to expand the gaps within the sample and promotes the secondary cracking of char [43]. This effect also explains the negative synergistic impact on char.

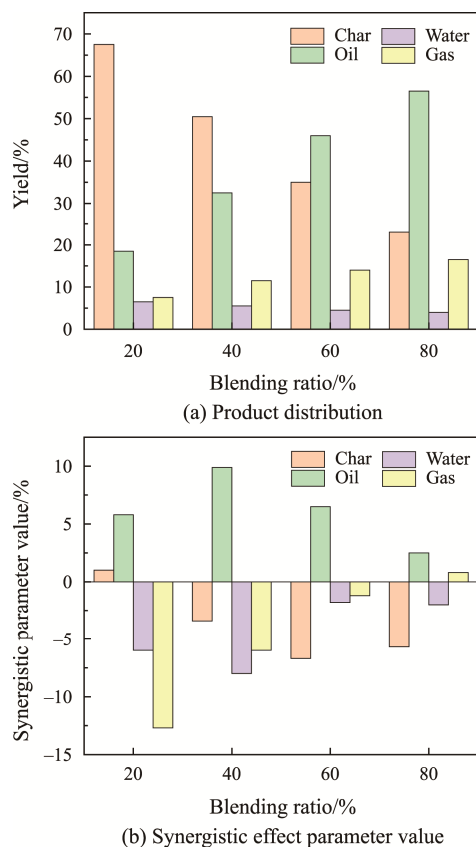


Fig. 4 Co-pyrolysis of bituminous coal and waste plastic

Fig. 5 illustrates the effect of the blending ratio on the distribution of co-pyrolysis products and the synergistic effect parameter values for lignitous coal and waste plastic under the same conditions as those shown in Fig. 4. As depicted in Fig. 5(a), the distribution of co-pyrolysis products for lignitous coal and waste plastic follows a pattern similar to that of bituminous coal and waste plastic. As the blending ratio increases, the yields of char and pyrolysis water gradually decrease, while

the yield of tar increases. The yield of pyrolysis gas initially decreases and then increases. Fig. 5(b) shows that the addition of waste plastic to lignitous coal has a positive synergistic effect on tar formation. The synergistic effect parameter value increases and then decreases with the blending ratio increasing, reaching a maximum value of 8.60% at a blending ratio of 40%.

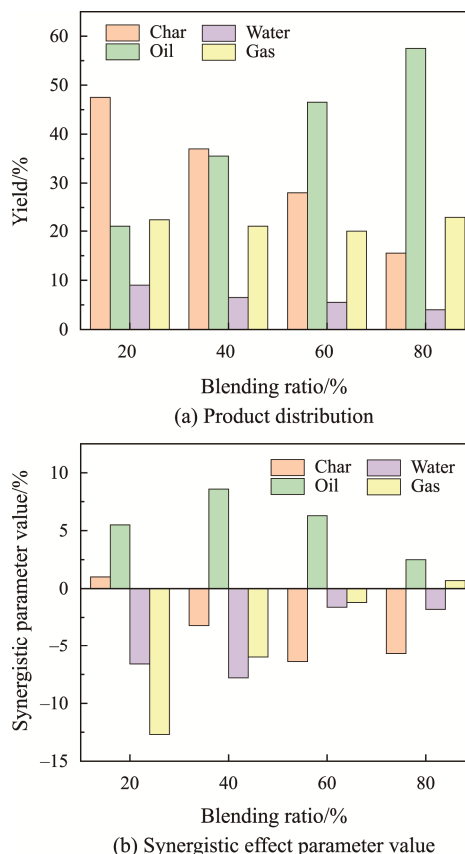


Fig. 5 Co-pyrolysis of lignitous coal and waste plastic

By comparing Figs. 4 and 5, it can be observed that under the same plastic blending ratio, as lignitous coal possesses higher H/C ratio, volatile matter, and moisture content, the co-pyrolysis yields of tar, water, and gas are all higher than those of bituminous coal. Meanwhile, its lower fixed carbon content significantly reduces the char yield. This is attributed to the fact that bituminous coal has a higher degree of coalification, and its intermolecular structure is more compact, making it less susceptible to cracking. Additionally, since its initial pyrolysis temperature and temperature range are much higher than those of plastic, there are fewer chances for the interaction among free radicals during the co-pyrolysis process [14]. Given the lower tar yield of bituminous coal, the increase in tar yield resulting from plastic is relatively higher, leading to a higher maximum synergistic parameter value than that of lignitous coal.

3.3 Optimization algorithms of the model

A Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) are utilized to optimize the BP neural network prediction model, with each prediction repeated three times and the average value taken as the final prediction error. The training set comprises 53 datasets, including Hebi lean coal mixed with waste PS and PE, Heshgewula lignite mixed with waste LDPE and PET, Huolin River lignite and Shenmu bituminous coal mixed with waste PE and PP, and bituminous coal and lignitous coal mixed with waste plastic. The testing set includes six datasets, featuring new conditions for Hebi lean coal mixed with waste PS and PE, and Pingshuo coal mixed with waste WPE. The hidden layer employs the tansig function, while the output layer uses the purelin function.

By comparing the prediction errors for models with various numbers of hidden layer nodes and learning rates, the optimal topology for the BP neural network model is determined to be 14-8-1, with a learning rate of 0.03. For the GA-BP model, the number of generations is set to 50, using the arithmetic Xover function for crossover and the non-UnifMutation function for mutation. The optimal topology for this model is found to be 14-7-1, with a learning rate of 0.01. For the PSO-BP model, the swarm size is set to 5, with an update frequency of 30; learning factors c_1 and c_2 both set to 4.494, and position and velocity changes (X and V) constrained within the range of $[-1,1]$ to prevent particles from exiting the search space. The optimal topology for the PSO-BP model is 14-11-1, with a learning rate of 0.01.

The uncertainty analysis of neural network models is a crucial means to ensure the reliability and robustness of the models. By analyzing the sources of uncertainty and employing appropriate optimization methods, the performance and generalization capabilities of the models can be effectively enhanced. In this study, methods including utilizing the most recent and inclusive datasets, data preprocessing, optimizing the neural network structure, and identifying the best parameters of optimization algorithms are employed to assess and reduce the uncertainty arising from data errors, model structure, and the coupling of optimization algorithms as much as possible.

The three models with optimal topologies and learning rates are then used to predict the tar yield for the database conditions, as illustrated in Fig. 6. This figure compares the predicted values with the experimental values. From Fig. 6(a), 6(c), and 6(e), it can be observed that for both training and testing samples, most data points fall within $\pm 10\%$ of the diagonal line, indicating a satisfactory level of consistency in tar yield prediction for all three models. Additionally, Fig. 6(b), 6(d), and 6(f) demonstrate that three models all exhibit strong predictive capabilities for tar yield.

The errors and performance of the three models are comprehensively evaluated by MAE, RMSE, MAPE, and R^2 . The larger the R^2 is, the smaller the MAE, RMSE and MAPE are, indicating that the more accurate the training model is, the better the performance is. The maximum errors of the predicted data $K_{\max}$ and average error $\bar{K}$ are also used to reflect the model's prediction performance more clearly using the following calculation formulas:

$$K_{\max} = \left(\frac{H_i - H_{pi}}{H_{pi}} \right)_{\max} \times 100\% \quad (18)$$

$$\bar{K} = \frac{1}{n} \left(\sum_{i=1}^n \frac{|H_i - H_{pi}|}{H_{pi}} \right) \times 100\% \quad (19)$$

Table 3 lists the prediction errors for the BP, GA-BP, and PSO-BP models. The data preprocessing leads to the extremely small MAE and RMSE. The results indicate that both GA and PSO have addressed some of the limitations of the BP model, leading to reduced errors and improved model performance. Notably, the R^2 of the GA-BP model is 0.9132, and the MAPE is 17.34%. Ahmad et al. [23] used LGBM regression model to predict biodiesel yield, R^2 is 0.9405, MAPE is 13.6%, and Alabdrabalnabi et al. [26] used dense neural network model to predict biochar and bio-oil yields from the co-pyrolysis of biomass and plastics, R^2 is 0.86 and 0.96, respectively. The comparison shows that the GA-BP model in this study has comparable predictive capacity in predicting tar yield of co-pyrolysis.

3.4 Verification of the optimal model

The previously established GA-BP neural network model has been used to rapidly predict the tar yield for 20 new conditions, including Hebi lean coal mixed with PE and PS, as well as Pingshuo coal mixed with PE.

3.4.1 Prediction of new operating conditions for existing coal types

Fig. 7 displays the prediction results of the GA-BP model for Hebi lean coal mixed with waste PE at temperatures of 475°C, 500°C, 525°C, and 550°C. The results indicate that, with the exception of 475°C, the predicted values aligns well with the actual values, maintaining a relative error within $(-20\%, 20\%)$. The average relative error of GBR is 18.15%, and XGB regression is 17.83% [23], which is slightly lower than that of the model in this study, showing that the GA-BP model has acceptable predictive capacity for engineering applications.

The excessive relative error observed at 475°C may be attributed to several factors, and reducing the error can be achieved in several ways. First, the available database (with data points fewer than 500) is too small for machine learning. The accuracy and precision of the

prediction model shall be improved by utilizing the most recent, high-quality, and well-structured datasets. We should expand the database as new data appear in the literature. On the other hand, the experimental data themselves are subjected to experimental uncertainty, which is on the order of several percent. Experimental

uncertainty detracts from model accuracy, and reducing the inherent experimental error can reduce the prediction errors. Second, errors can be reduced by better data preprocessing (i.e. data enhancement [27]) and proper partitioning of data sets. Using only one training set for model training can easily lead to limited generalization

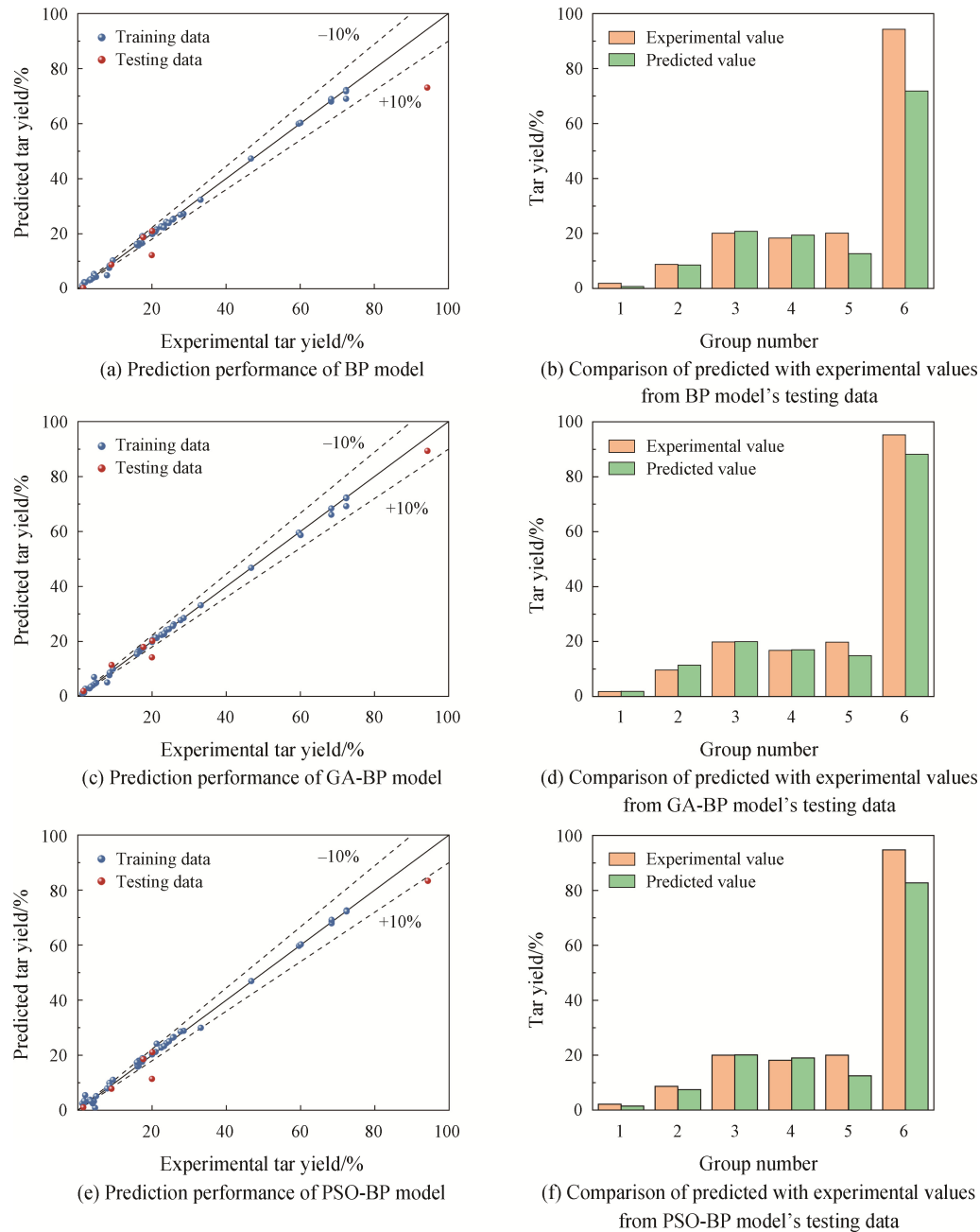


Fig. 6 Comparison of tar yield prediction and experimental values for the three models

Table 3 Prediction errors of tar yield for different models

Model types	R^2	MAE	RMSE	MAPE/%	$K_{max}/\%$	$\bar{K}/\%$
BP	0.8275	0.0143	0.0227	29.47	77.75	25.50
GA-BP	0.9132	0.0101	0.0147	17.34	28.93	14.98
PSO-BP	0.8767	0.0108	0.0170	21.49	43.04	18.27

ability and stability of the model. Therefore, using multiple different training and testing sets can improve the robustness, generalization ability, and stability of the model [29]. Third, combining newly developed prediction and regression models with the model proposed and improving the optimization algorithm can achieve higher performance and more accurate prediction results.

Fig. 8 presents the prediction results of the GA-BP model for Hebi lean coal mixed with waste PS at

temperatures of 475°C, 500°C, 525°C, and 550°C. The results indicate that, except for 500°C, the predicted values are consistent with the actual values in an engineering-acceptable manner, with the relative error maintained within (−20%, 20%).

3.4.2 Prediction of new operating conditions for new coal types

Fig. 9 illustrates the prediction results of the GA-BP model for Pingshuo coal mixed with waste WPE at

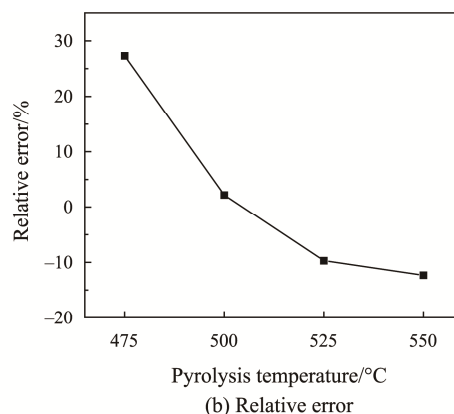
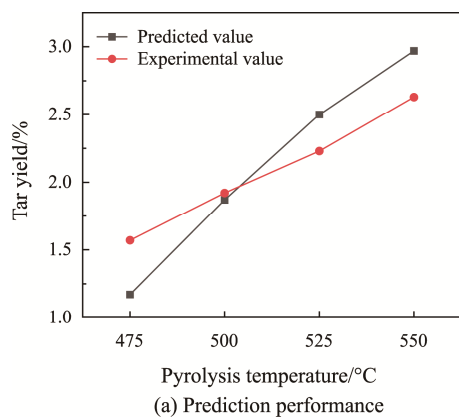


Fig. 7 Prediction of tar yield from the co-pyrolysis of Hebi lean coal and waste PE

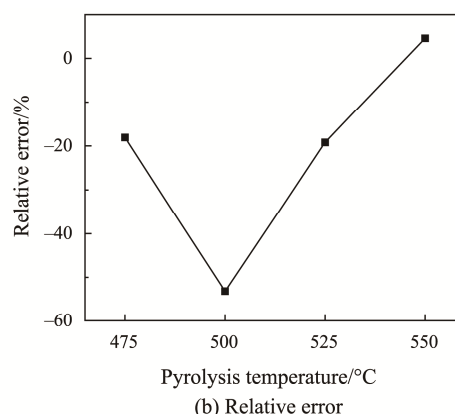
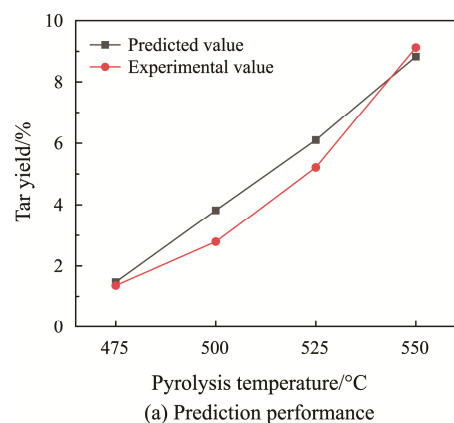


Fig. 8 Prediction of tar yield from the co-pyrolysis of Hebi lean coal and waste PS

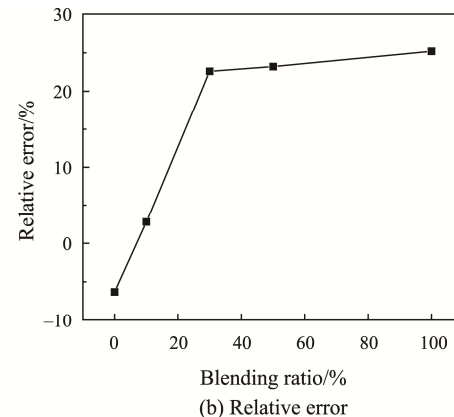
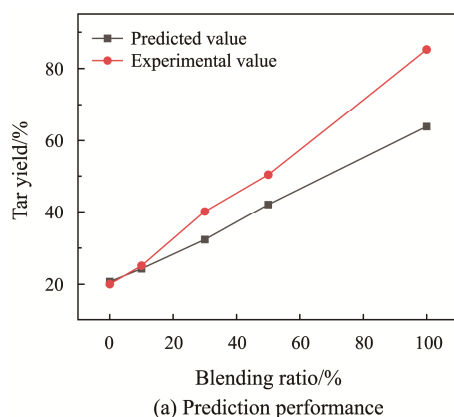


Fig. 9 Prediction of tar yield from the co-pyrolysis of Pingshuo coal and waste WPE

blending ratios of 0%, 10%, 30%, 50%, and 100%. The results indicate that the predicted values align with the actual values in an engineering-acceptable manner, with the relative error maintained within (−10%, 30%). This demonstrates the utility of the model in predicting the tar yield from the co-pyrolysis of coal and waste plastic.

Fig. 10 depicts the prediction results of the GA-BP model for Pingshuo coal mixed with waste BPE at blending ratios of 0%, 10%, 30%, 50%, and 100%. The results indicate that the predicted values align with the actual values in an acceptable manner, with the relative error maintained within (−20%, 10%).

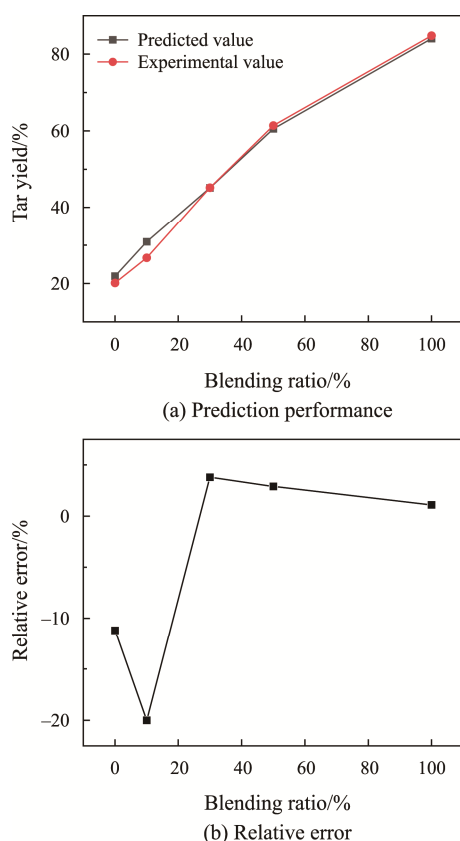


Fig. 10 Prediction of tar yield from the co-pyrolysis of Pingshuo coal and waste BPE

4. Conclusions

(1) The addition of waste plastic significantly enhances the tar yield from bituminous coal and lignitous coal. The synergistic effect parameter values initially increase and then decrease with an increase in the waste plastic blending ratio, reaching maximum values of 9.89% and 8.60% at the same blending ratio of 40%.

(2) The continuous increase in the mixing ratio facilitates the formation of a liquid phase on the surface of coal particles during pyrolysis, thus inhibiting the release of tar and resulting in a gradual decrease in the

positive synergistic effect of waste plastic on tar.

(3) The GA-BP model achieves a 10.52% reduction in the average error under training conditions, and its R^2 is 0.9132, showing that it has comparable predictive capacity. When used to predict tar yield under new conditions in the database, the model maintains a relative error within (−20%, 30%), indicating relatively good accuracy and practicality.

(4) The excessive relative error can be reduced by utilizing high-quality datasets including new literature data and more accurate experimental data, better data preprocessing and combining newly developed prediction and regression models with the model proposed in this study.

In this study, the co-pyrolysis characteristics of coal and waste plastic have been examined, and a rapid prediction model for tar yield has been constructed, optimized and verified. The strength of this study is the quantification of the synergistic effects of waste plastics on coal pyrolysis, and the proposed model enables rapid and convenient prediction of co-pyrolysis tar yield, significantly saving time and cost. The limitation is that the inherent errors in experimental data and the constraints of model optimization methods result in relatively large prediction errors still. In the future, after applying the error-reduction optimization strategies proposed to the model, it is believed that it will have better predictive performance and generalizability. The rapid prediction model proposed in this study can also be used for predicting the yields of co-pyrolysis gas and char, which aids in the optimization and regulation of co-pyrolysis processes aimed at achieving the best oil and gas yields.

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Conflict of Interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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